LRTuckerRep: Low-rank Tucker Representation Model for Multi-dimensional Data Completion

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2025 8th International Conference on Digital Medicine and Image Processing
Tokyo University, November 29, 2025

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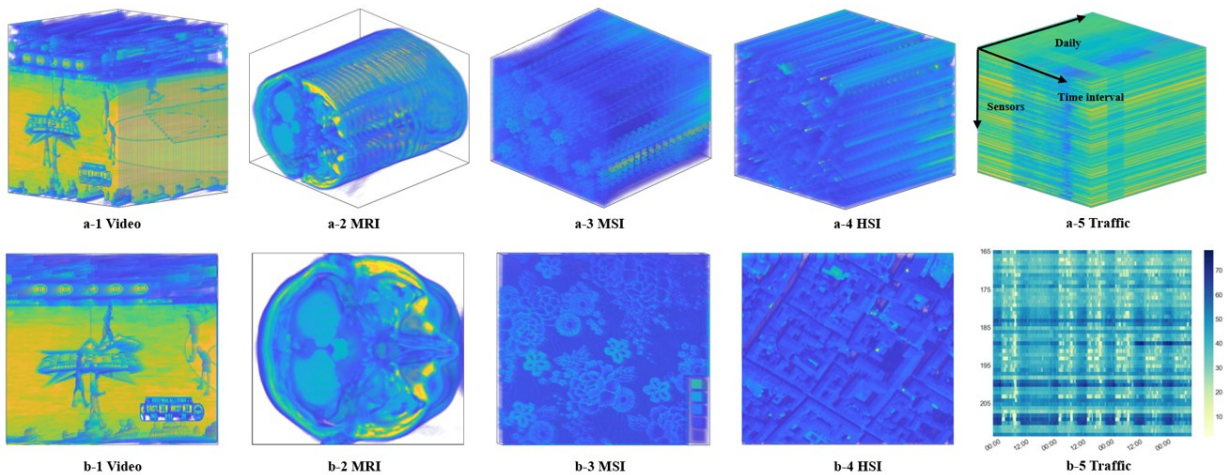
Outline

1. Introduction
2. The Proposed LRTuckerRep Model
3. Optimization Algorithm
4. Experimental Validation
5. Conclusion

Multi-dimensional Data: Tensors

Tensors provide a natural representation for multi-dimensional data, preserving higher-order patterns (**global correlations and local similarities**).

- Images (height, width, channel)
- Videos (height, width, channel, time)
- Medical Imaging (e.g., fMRI)
- Spatialtemporal Data (e.g., traffic)



Visualizing tensor data from imaging domains.

The Problem: Corrupted Tensor

Real-world data is often incomplete due to sensor failures, occlusions, or communication errors. This severely degrades the performance of downstream tasks!

Tensor Completion Task

Goal: Recover a complete tensor $\mathcal{X}$ from its partially observed version $\mathcal{T}$.

From a Bayesian perspective, this is a Maximum a Posteriori (MAP) estimation:

Formal Definition

$$\begin{aligned}\hat{\mathcal{X}} &= \arg\max_{\mathcal{X}} P(\mathcal{X} \mid \mathcal{T}) = \arg\max_{\mathcal{X}} \frac{P(\mathcal{T} \mid \mathcal{X})P(\mathcal{X})}{P(\mathcal{T})} \\ &= \arg\max_{\mathcal{X}} \{\log P(\mathcal{T} \mid \mathcal{X}) + \log P(\mathcal{X})\} = \arg\min_{\mathcal{X}} \ \mathcal{R}(\mathcal{X}), \quad \text{s.t., } \mathcal{X}_\Omega = \mathcal{T}_\Omega,\end{aligned}$$

- Ω: The set of indices of observed entries.
- $\mathcal{R}(\mathcal{X})$: A regularization term encoding **prior knowledge** about the tensor's intrinsic structure.

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Insight: The key to successful tensor completion (TC) lies in the design of an effective prior $\mathcal{R}(\mathcal{X})$.

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Two Key Structural Priors

Real-world tensors typically exhibit two complementary properties:

Global Correlation (Low-rankness)

- Captures high correlation across different modes.
- Methods: Nuclear Norm Minimization, Low-rank Tensor Factorization

→ **Pros**: Models global structure.

→ **Cons**: High computational cost, may disrupt local details.

Local Similarity (Smoothness)

- Adjacent elements tend to be similar.
- Methods: Total Variation (TV), Laplacian Regularization.

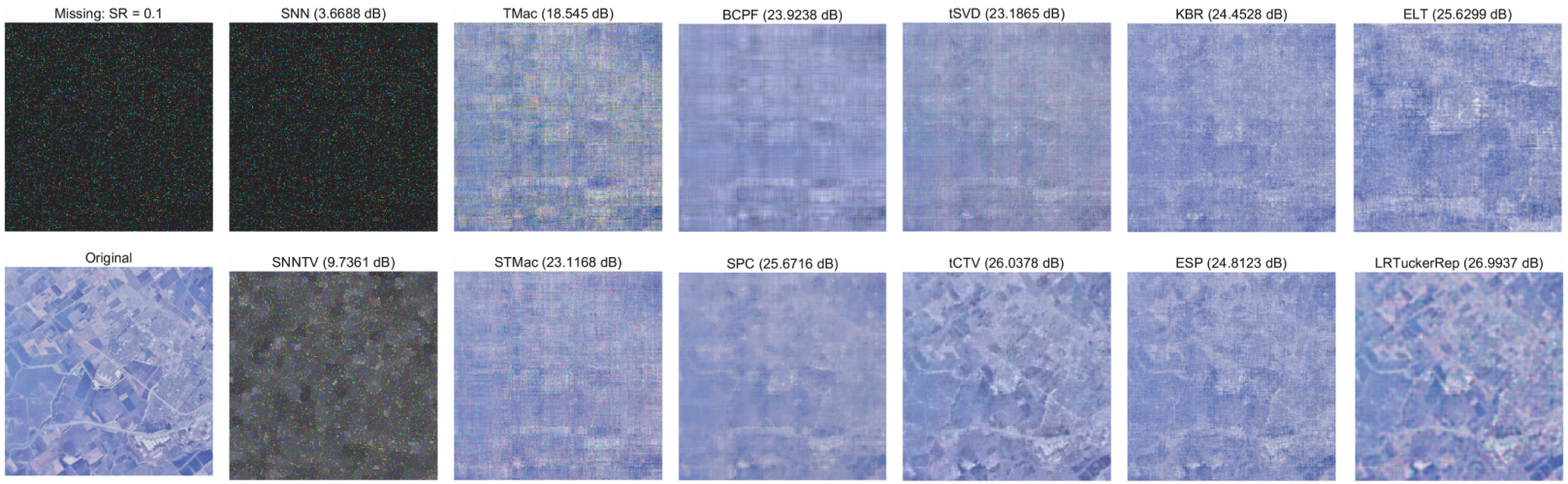
→ **Pros**: Preserves edges and fine details.

→ **Cons**: Requires manual parameter tuning, poor generalization.

The Core Challenge: Unifying Priors Effectively

The Dilemma

Simple combinations of low-rank and smooth priors often fail. The performance heavily depends on manually-tuned hyperparameters.

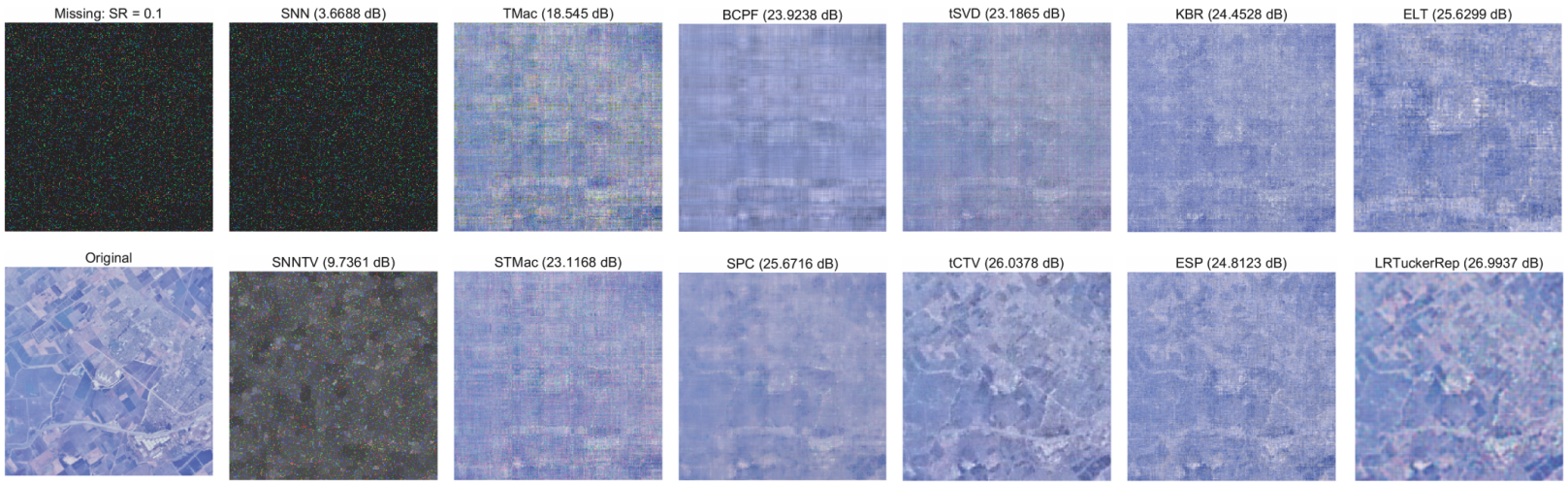


Top: Low-rank only methods. Bottom: Joint prior methods. Our proposal (LRTuckerRep) shows significant improvement.

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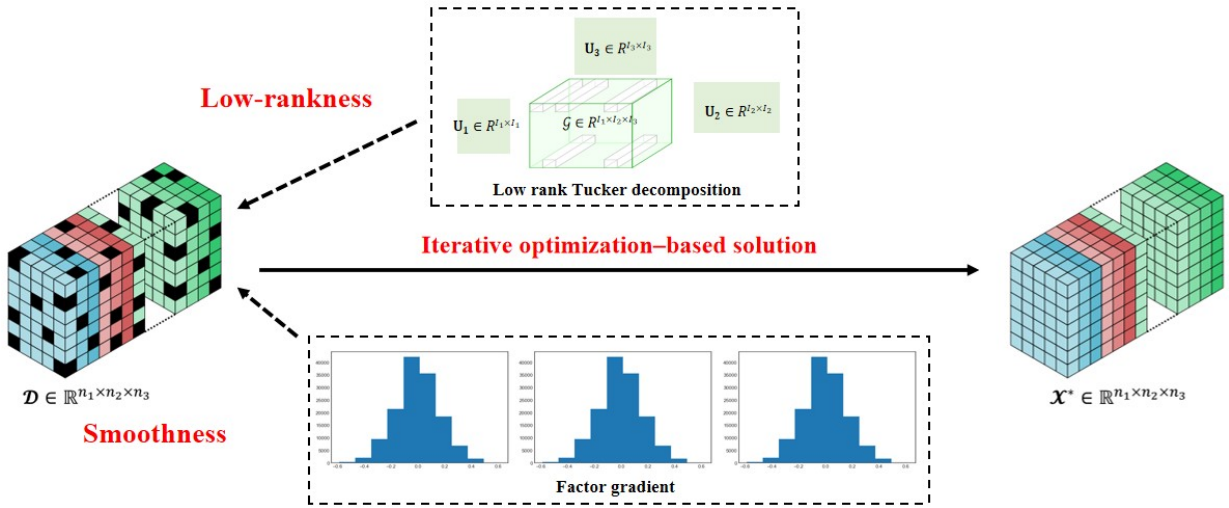
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Research Questions

- How to design a unified model that adaptively balances global and local priors?
- How to eliminate the need for hyperparameter tuning?
- How to develop an efficient, provably convergent algorithm for a nonconvex model?

Our Core Idea: A Unified Tucker Representation

We propose to unify low-rankness and smoothness priors **within the Tucker decomposition framework**. Instead of regularizing $\mathcal{X}$ directly, we regularize its Tucker components $\mathcal{G}$ and $\{\mathbf{U}_n\}$.



A Unified Low-rank Tucker Representation.

Tucker Decomposition Recap

$\mathcal{X} = \mathcal{G} \times_1 \mathbf{U}_1 \times_2 \mathbf{U}_2 \cdots \times_N \mathbf{U}_N$

- $\mathcal{G}$: The core tensor, capturing interactions between modes.
- $\{\mathbf{U}_n\}$: Factor matrices, representing the principal components in each mode.

The LRTuckerRep Model Formulation

Objective Function

$$\min_{\mathcal{G},\{\mathbf{U}_n\}} \underbrace{(1-\alpha)\sum_{n=1}^{N}\omega_n\|\mathbf{U}_n\|_* + \alpha\|\mathcal{G}\|_1}_{\text{Novel Low-Rank Prior}} + \underbrace{\sum_{n\in\Gamma}\frac{\beta_n}{2}\operatorname{tr}(\mathbf{U}_n^T\mathbf{L}_n\mathbf{U}_n)}_{\text{Adaptive Smoothness Prior}}$$

A Tighter Low-rank Tucker Measure

- **Sparse Core** (ℓ_1-norm on $\mathcal{G}$): Enforces a stronger low rank measure.
- **Low-Rank Factors** (Weighted Nuclear Norm on $\mathbf{U}_n$): Promotes low-rankness in each mode.

Adaptive Smoothness Regularization

- **Factor Smoothness**: Laplacian regularization $\mathbf{L}_n$ on factor matrices captures local similarity.
- **Parameter-Free**: Weights ω_n and β_n are determined adaptively from the data.

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A Tighter Low-rank Tucker Measure

Complementary Regularization

$$\mathcal{R}(\mathcal{G}, \{\mathbf{U}_n\}) = (1-\alpha)\sum_{n=1}^{N} \omega_n \|\mathbf{U}_n\|_* + \alpha\|\mathcal{G}\|_1, \quad 0 < \alpha < 1$$

- **Nuclear Norm on Factor Matrices:** $\|\mathbf{U}_n\|_* = \sum_i \sigma_i(\mathbf{U}_n)$ penalizes singular values, controlling the **dimensionality of latent subspace** in each mode and capturing **global correlations**.
- ℓ_1**-norm on Core Tensor:** $\|\mathcal{G}\|_1 = \sum_{i_1,\ldots,i_N} |\mathcal{G}_{i_1\cdots i_N}|$ penalizes coefficients in latent representation, controlling how basis vectors are **combined** and capturing **interaction structure**.

Our combined regularizer is tighter because:

- Simultaneously penalizes both subspace dimensions and coefficient sparsity
- ℓ_1-norm on $\mathcal{G}$ reduces effective rank of $\mathbf{G}_{(n)}$
- Parameter α enables adaptive balancing

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The Adaptive Mechanism

The key to our model's robustness is the **adaptive hyperparameter tuning**.

Adaptive Low-rankness Weight ω_n

$\omega_n \propto \prod_{i \neq n} \frac{1}{\sum_j \sigma_j(\mathbf{U}_i)}$

- Balances the contribution of each factor matrix's nuclear norm.
- Provides a novel interpretation of low-rank based on tensor sparsity.

Adaptive Smoothness Weight β_n

$\beta_n \propto \frac{\sigma_1(\mathbf{X}_{(n)})}{\sigma_1(\mathbf{L}_n)}$

- Balances the scale of the data energy ($\sigma_1(\mathbf{X}_{(n)})$) and the smoothness operator energy ($\sigma_1(\mathbf{L}_n)$).
- Adaptively adjusts the strength of smoothness regularization for each mode.

The Optimization Challenge

Our objective function is:

- **Non-convex**: Due to the coupled multiplicative terms in the Tucker representation.
- **Non-smooth**: Due to the ℓ_1 and nuclear norms.
- **Multi-block**: Involves multiple blocks of variables $(\mathcal{G}, \mathbf{U}_1, \ldots, \mathbf{U}_N)$.

How to solve it efficiently with theoretical guarantees?

We propose algorithms based on the Proximal Alternating Linearized Minimization (PALM) and Proximal Alternating Direction Multiplier (ProADM) framework.

Strategy: Iteratively solve for each block of variables while keeping others fixed. For each subproblem, we linearize the smooth part of the objective and solve the resulting proximal problem.
Acceleration: We incorporate a FISTA-like extrapolation step $(\tilde{\mathcal{G}}^k, \tilde{\mathbf{U}}_n^k)$ to accelerate convergence.

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The PALM Algorithm

PALM-based Solver

$$\min_{\mathcal{G},\{\mathbf{U}_n\},\mathcal{X}} (1-\alpha)\sum_{n=1}^{N}\omega_n\|\mathbf{U}_n\|_* + \alpha\|\mathcal{G}\|_1 + \sum_{n\in\Gamma}\frac{\beta_n}{2}\operatorname{tr}\left(\mathbf{U}_n^{\mathrm{T}}\mathbf{L}_n\mathbf{U}_n\right) + \frac{1}{2}\left\|\mathcal{G}\times_{n=1}^{N}\mathbf{U}_n - \mathcal{X}\right\|_{\mathrm{F}}^2$$

- Update Core Tensor $\mathcal{G}$ via **Soft-Thresholding operator**.

$$\mathcal{G}^{k+1} = \mathcal{S}_{\tau_g}(\cdot) = \mathcal{S}_{\frac{\alpha}{L_{\mathcal{G}}^k}}\left(\tilde{\mathcal{G}}^k - \frac{1}{L_{\mathcal{G}}^k}\nabla_{\mathcal{G}}H\left(\tilde{\mathcal{G}}^k\right)\right)$$

- Update Factor Matrices $\mathbf{U}_n$ via **Singular Value Shrinkage operator**.

$$\mathbf{U}_n^{k+1} = \mathcal{D}_{\tau_u}(\cdot) = \mathcal{D}_{\frac{(1-\alpha)\omega_n}{L_{\mathbf{U}_n}^k}}\left(\tilde{\mathbf{U}}_n^k - \frac{1}{L_{\mathbf{U}_n}^k}\nabla_{\mathbf{U}_n}H\left(\tilde{\mathbf{U}}_n^k\right)\right)$$

- Update tensor $\mathcal{X}^{k+1} = \mathcal{T}_\Omega + \left(\mathcal{G}^{k+1}\times_{n=1}^{N}\mathbf{U}_n^{k+1}\right)_{\bar{\Omega}}$

Convergence Guarantee

Despite the non-convexity, we provide a rigorous theoretical guarantee.

Theorem (Global Convergence)

Let $\{\Theta^k\} = \{\mathcal{G}^k, \{\mathbf{U}_n^k\}\}$ *be the sequence generated by our PALM-based algorithm. Then,* $\{\Theta^k\}$ *is a Cauchy sequence and is guaranteed to converge to a critical point* $\hat{\Theta}$ *of the objective function.*

Proof Sketch

The proof relies on establishing that our objective function satisfies the **Kurdyka-Lojasiewicz (KL) property**, a key condition for ensuring convergence in non-convex optimization. We show that:

1. The objective function value is guaranteed to decrease at each iteration.
2. The sequence of iterates has a finite length and is therefore bounded.
3. The sufficient descent and continuity properties hold, which, combined with the KL property, proves global convergence.

The ProADM Algorithm

ProADM-based Solver

$$\min_{\mathcal{G},\{\mathbf{U}_n\},\mathcal{X},\mathcal{P}^{\mathcal{X}},\mathcal{P}^{\Omega},\mu} (1-\alpha)\sum_{n=1}^{N}\omega_n\|\mathbf{U}_n\|_* + \alpha\|\mathcal{G}\|_1 + \sum_{n\in\Gamma}\frac{\beta_n}{2}\operatorname{tr}\left(\mathbf{U}_n^{\mathrm{T}}\mathbf{L}_n\mathbf{U}_n\right)$$
$$+\frac{\mu}{2}\left(\left\|\mathcal{X}-\mathcal{G}\times_{n=1}^{N}\mathbf{U}_n\right\|_{\mathrm{F}}^2+\left\|\mathcal{X}_{\Omega}-\mathcal{T}_{\Omega}\right\|_{\mathrm{F}}^2\right)$$
$$+\left\langle\mathcal{P}^{\mathcal{X}},\mathcal{X}-\mathcal{G}\times_{n=1}^{N}\mathbf{U}_n\right\rangle+\left\langle\mathcal{P}^{\Omega},\mathcal{X}_{\Omega}-\mathcal{T}_{\Omega}\right\rangle,$$

$$\mathcal{G}^{k+1}\approx\mathcal{S}_{\frac{\alpha}{L_{\mathcal{G}}^k}}\left(\mathcal{G}^k-\frac{1}{L_{\mathcal{G}}^k}\nabla_{\mathcal{G}}f\left(\mathcal{G}^k\right)\right),\quad \mathbf{U}_n^{k+1}\approx\mathcal{D}_{\frac{(1-\alpha)\omega_n}{L_{\mathbf{U}_n}^k}}\left(\mathbf{U}_n^k-\frac{1}{L_{\mathbf{U}_n}^k}\nabla_{\mathbf{U}_n}f\left(\mathbf{U}_n^k\right)\right)$$

$$\mathcal{X}_{\Omega}^{k+1}=\frac{1}{2}\left(\mathcal{G}^{k+1}\times_{n=1}^{N}\mathbf{U}_n^{k+1}-\frac{\mathcal{P}_k^{\mathcal{X}}}{\mu_k}+\mathcal{T}-\frac{\mathcal{P}_k^{\Omega}}{\mu_k}\right)_{\Omega},$$
$$\mathcal{X}_{\bar{\Omega}}^{k+1}=\left(\mathcal{G}^{k+1}\times_{n=1}^{N}\mathbf{U}_n^{k+1}-\frac{\mathcal{P}_k^{\mathcal{X}}}{\mu_k}\right)_{\bar{\Omega}}.$$

Experimental Setup

Random Missing (RM) scenarios with varying sampling rates (1% to 95%)

Image Data Metrics: Higher values indicate better performance

MPSNR (Mean Peak Signal-to-Noise Ratio): $10 \cdot \log_{10} \frac{(\mathcal{X}_{\max})^2}{\|\hat{\mathcal{X}} - \mathcal{X}_{\text{true}}\|_{\text{F}}^2 / |\bar{\Omega}|}$

MSSIM (Mean Structural Similarity): $\frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$

Traffic Data Metrics: Lower values indicate better performance

NMAE (Normalized Mean Absolute Error): $\frac{\sum_{i=1}^{n} |y_i - \hat{y}_i|}{\sum_{i=1}^{n} |y_i|}$

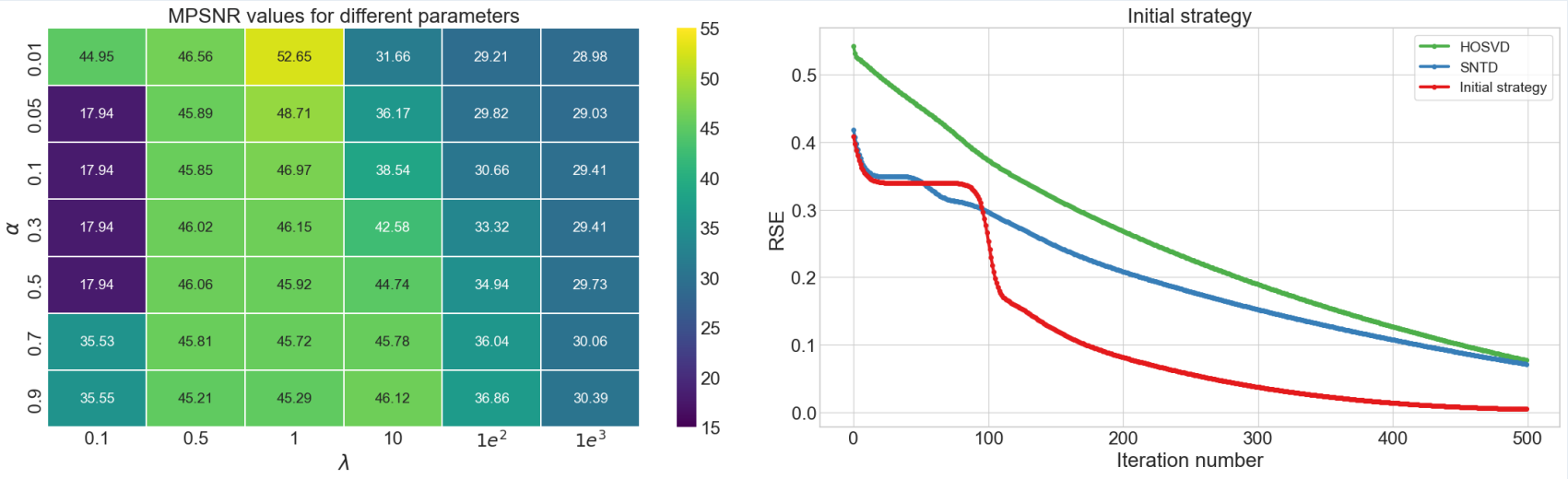
MAPE (Mean Absolute Percentage Error): $\frac{1}{n} \sum_{i=1}^{n} \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100$

Baselines: We compare against a comprehensive set of tensor-guided model driven methods, including:

- *Low-rank only*: HaLRTC, TMac, BCPF, KBR, tSVD
- *Joint prior*: SNNTV, STMac, SPC, ESP, tCTV, SparsityTD, LATC, stTT

Synthetic Data: Parameter Analysis

Parameter Sensitivity Analysis

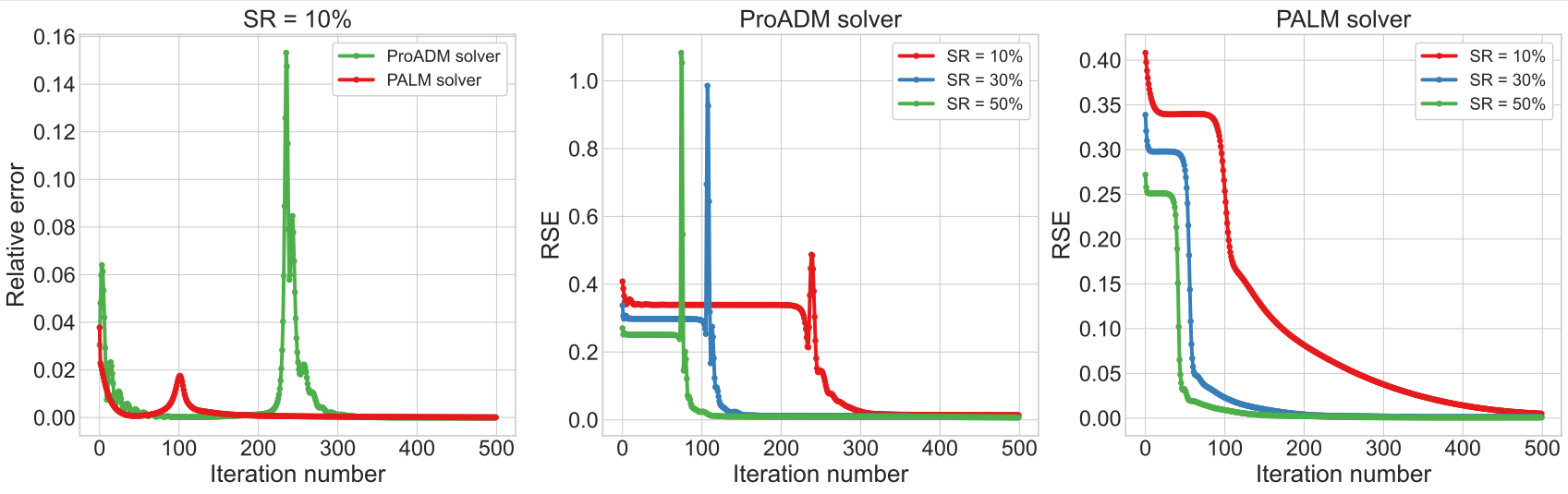


Key Findings:

- Model is robust to penalty parameter λ
- Highly sensitive to balance parameter α
- Optimal configuration: $\alpha = 0.01$, $\lambda = 1$
- Initial strategy effectively reduces errors

Synthetic Data: Convergence

Numerical Convergence



Convergence Verification:

- Relative error converges to zero
- RSE curves show monotonic decrease
- Both PALM and ProADM algorithms are numerically stable
- Theoretical convergence guarantees validated

Image Inpainting: Quantitative Comparison

MPSNR Performance Comparison (5% Sampling Rate)

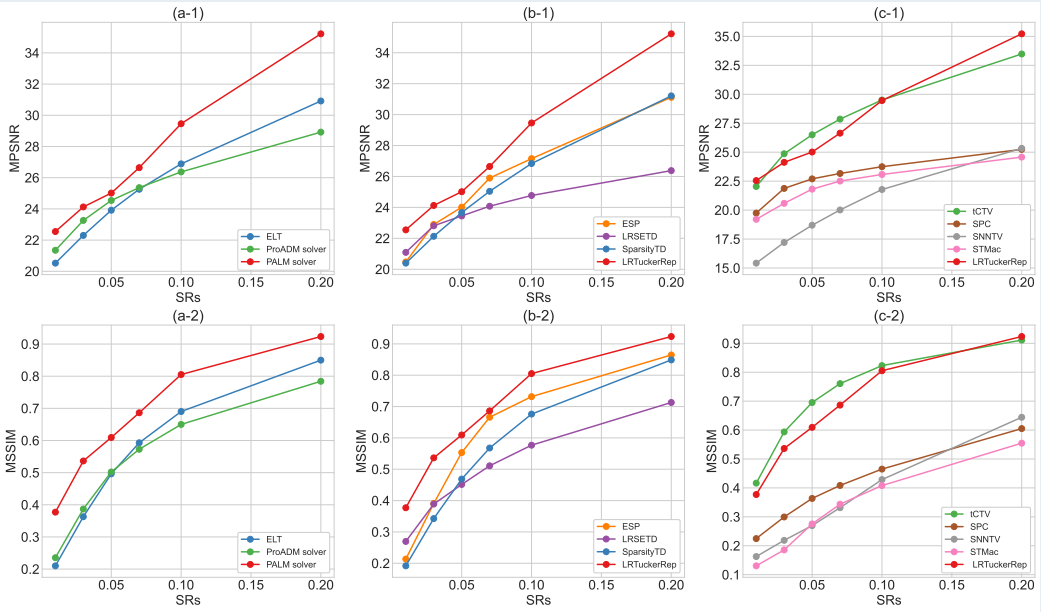
Methods	RGB	MRI	MSI
LRTuckerRep	**26.99**	**26.97**	25.05
tCTV	24.24	26.88	**26.50**
SparsityTD	24.15	24.93	24.78
LRSETD	24.25	22.76	23.45
SPC	24.31	21.88	23.68
STMac	20.64	21.85	20.54

Key Observations

LRTuckerRep consistently achieves **top-tier performance** across different image modalities (RGB, MRI, MSI). The model demonstrates particular strength in natural images (RGB) and medical imaging (MRI), while maintaining competitive performance in multispectral imaging tasks.

Image Inpainting: Performance Curves

MSI Recovery

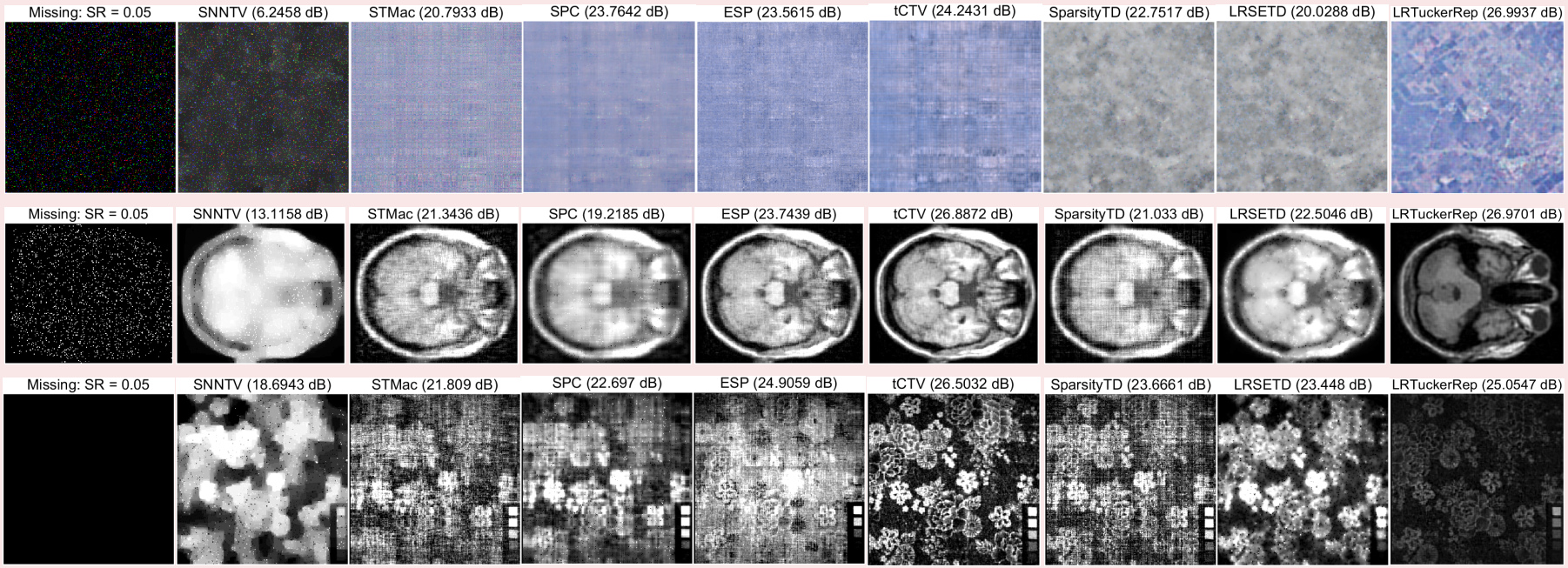


MSI 'Cloth' (256×256×31):

- Smoothness prior significantly improves performance
- PALM algorithm shows superior accuracy
- Robust performance across different sampling rates
- Outstanding performance at low sampling rates (<20%)

Image Inpainting: Robustness in Extreme Scenario

95% Missing Data Results

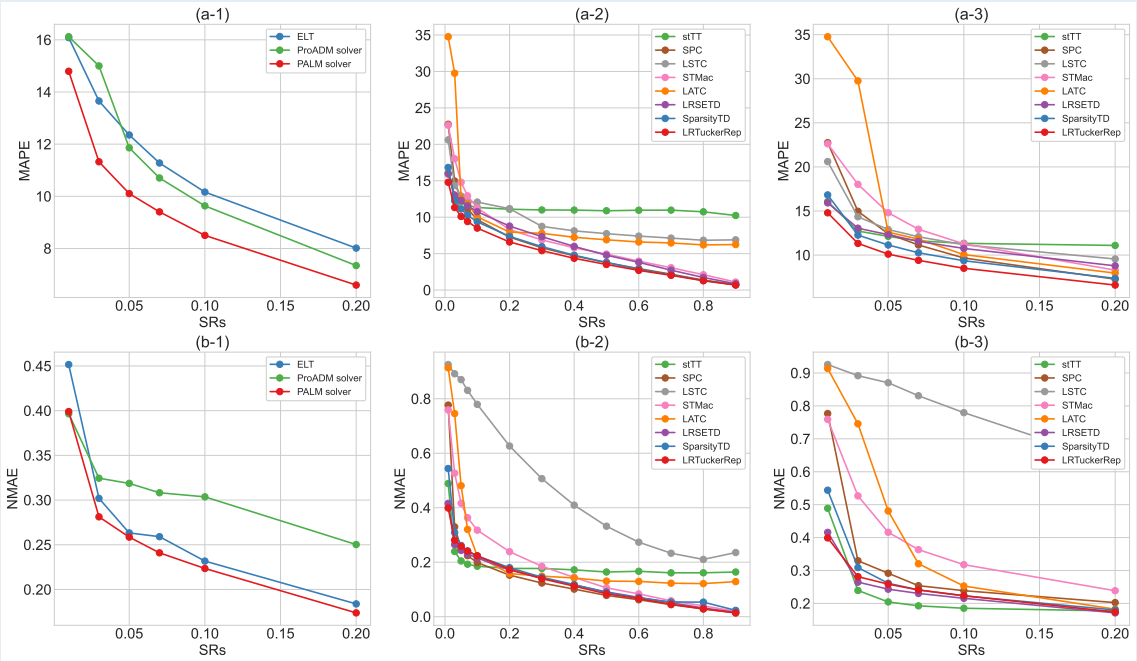


Even with **95% of data missing**, LRTuckerRep successfully recovers meaningful image structures and textures. The visual comparison clearly shows that our method produces results closest to the ground truth, significantly outperforming traditional approaches like tCTV and SparsityTD in challenging scenarios.

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Traffic Data Imputation: Spatiotemporal Performance

UTD (Urban Traffic)

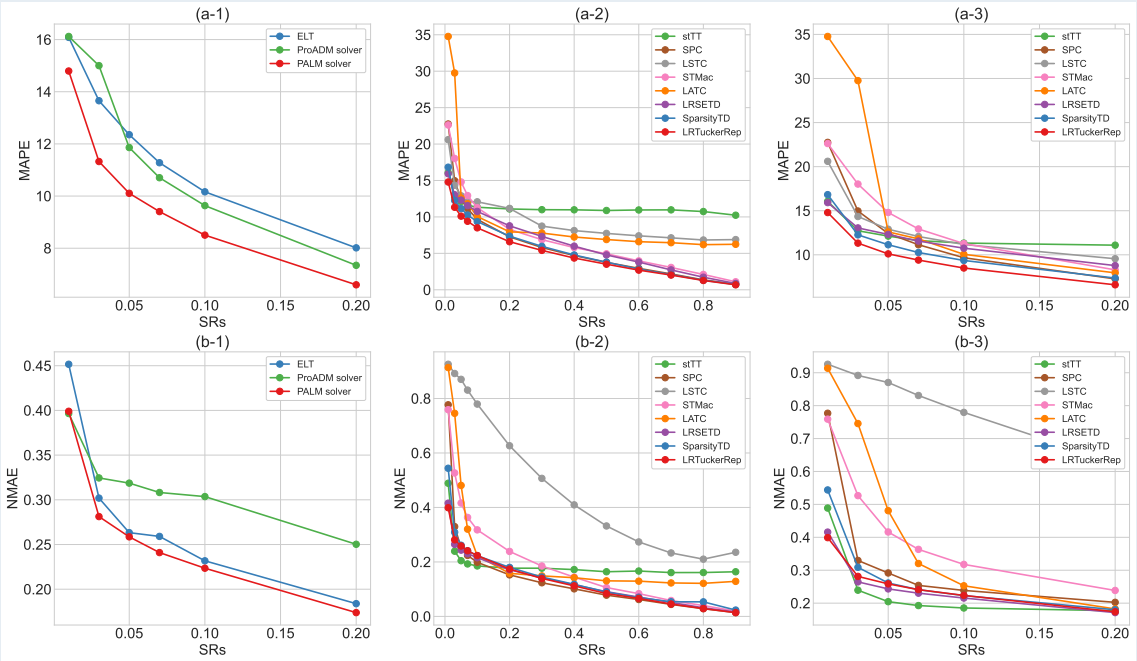


Guangzhou Urban Traffic (214×144×7):

- Factor gradients effectively capture spatiotemporal patterns
- Lowest MAPE and NMAE across all sampling rates
- Superior performance in urban traffic scenarios

Traffic Data Imputation: Spatiotemporal Performance

NTD (Internet Traffic)



Abilene Internet Traffic (121×288×7):

- Consistent superior performance across datasets
- Effective generalization to different traffic types
- Robust performance even at extreme missing rates

Summary of Experimental Results

Key Experimental Findings

Consistent Superior Performance: LRTuckerRep outperforms state-of-the-art methods across diverse applications including RGB images, medical imaging (MRI), multispectral data, and spatiotemporal traffic datasets.

Extreme Robustness: The model maintains excellent performance even with up to 95% missing data, demonstrating exceptional robustness in challenging real-world scenarios.

Self-Adaptive Mechanism: The incorporation of parameter-adaptive smoothness regularization eliminates manual hyperparameter tuning while significantly improving reconstruction quality.

Algorithmic Reliability: Both PALM and ProADM algorithms demonstrate numerical stability and convergence, providing theoretical guarantees.

LRTuckerRep provides a **robust, accurate, and generalizable** solution for multi-dimensional data completion, making it suitable for real-world applications.

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Take Home Messages

- A unified global low-rank and local smoothness priors within a Tucker representation framework is powerful for TC.
- **LRTuckerRep** features a **new low-rank measure** (sparse core + low-rank factors) and an **adaptive smoothness regularization**.
- **PALM-based solver** is an **efficient** and **convergence guaranteed** algorithm for solving Tucker-based models.

Future Work

- **Application Expansion**: Extend the model to other inverse problems like deblurring and denoising, and apply it to more domains such as computational biology and finance.
- **Algorithmic Improvement**: Develop stochastic or distributed optimization algorithms to handle large-scale tensor data.
- **Methodology Enhancement**: Investigate the combination of our model with deep learning, exploring deep unfolding networks or learning the transformation in a data-driven manner.

Thank You!

Questions?

Citation: Gong, W. and Yang, L. (2025). LRTuckerRep: Low-rank Tucker Representation Model for Multi-dimensional Data Completion. arXiv preprint arXiv:2508.03755.

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